QIAO Bowen · FYP UPDATE · 8 OCTOBER 2026

Historical OCR: learning and context

EXP-013–017: page selection, inference-time examples and the next linguistic research question.

Adaptation works; the role of context is still open

What we observed

Fine-tuning gives the largest gain.
Active selection has no sustained lead.
Text-only ICL is close to paired ICL.
LightOn fails under transferred chat setup.

What remains unresolved

Which language cues drive the gain?
Why do many individual lines worsen?
Can text adaptation approach image/text adaptation under matched conditions?

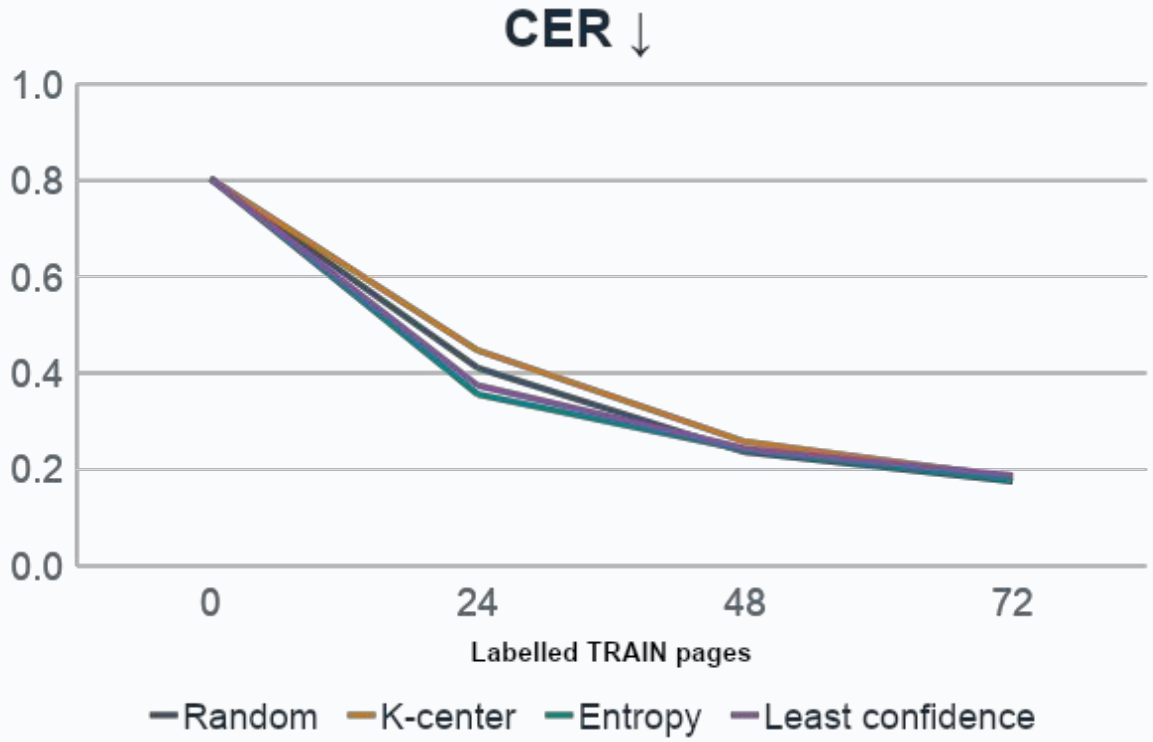
Recognition uses supplied line boxes; these results do not measure detection or final-test generalization.

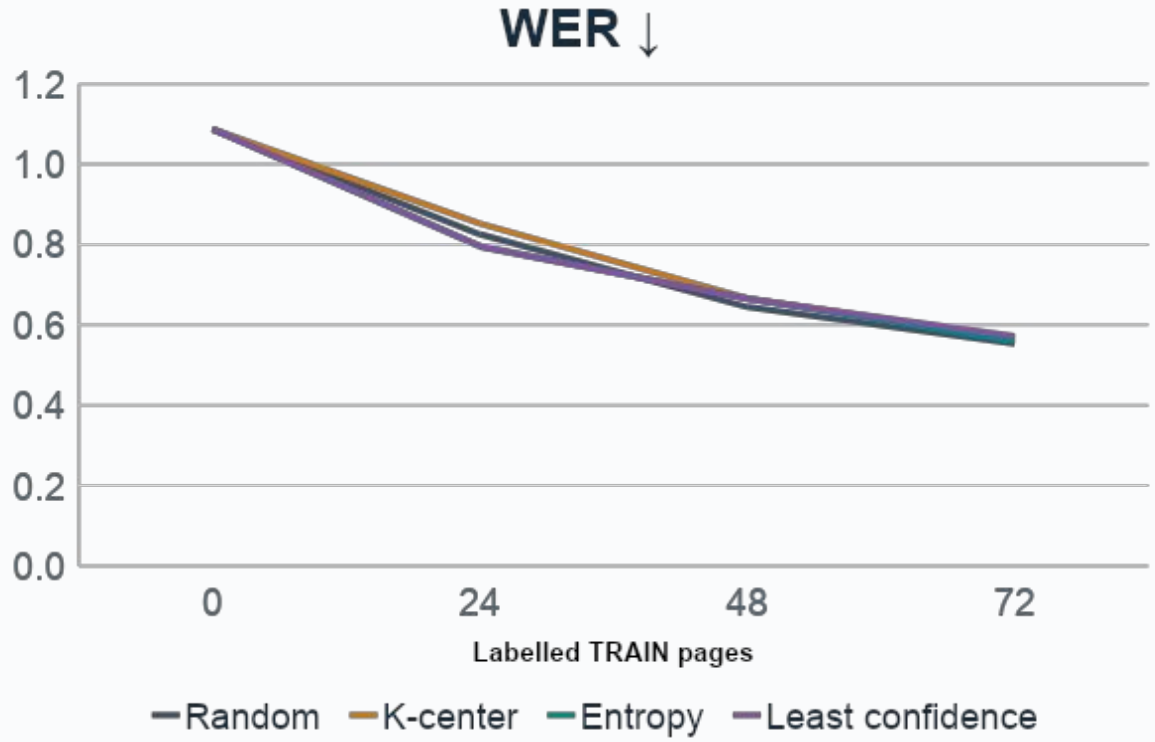
Next question

What linguistic information helps a VLM read historical handwriting,
and when does that information override the image?

Evidence: EXP-013–017 reader reports · one-bank, official-validation pilots

Uncertainty leads early; random leads later





At 24 pages, entropy leads random by 5.45 CER points; the lead disappears.

Independent first selections · fresh cumulative fits · same VAL42 / 861 lines

The advantage depends on the label budget

CER ↓ · historical page metric

Strategy	0*	24	48	72
Random	0.8036	0.4104	0.2357	0.1750
K-center	0.8036	0.4469	0.2574	0.1838
Entropy	0.8036	0.3559	0.2400	0.1781
Least conf.	0.8036	0.3740	0.2439	0.1881

WER ↓ · historical page metric

Strategy	0*	24	48	72
Random	1.0871	0.8254	0.6445	0.5529
K-center	1.0871	0.8523	0.6662	0.5664
Entropy	1.0871	0.7951	0.6665	0.5593
Least conf.	1.0871	0.7963	0.6650	0.5716

Entropy vs least confidence overlaps: 16/24 → 24/48 → 37/72 pages.

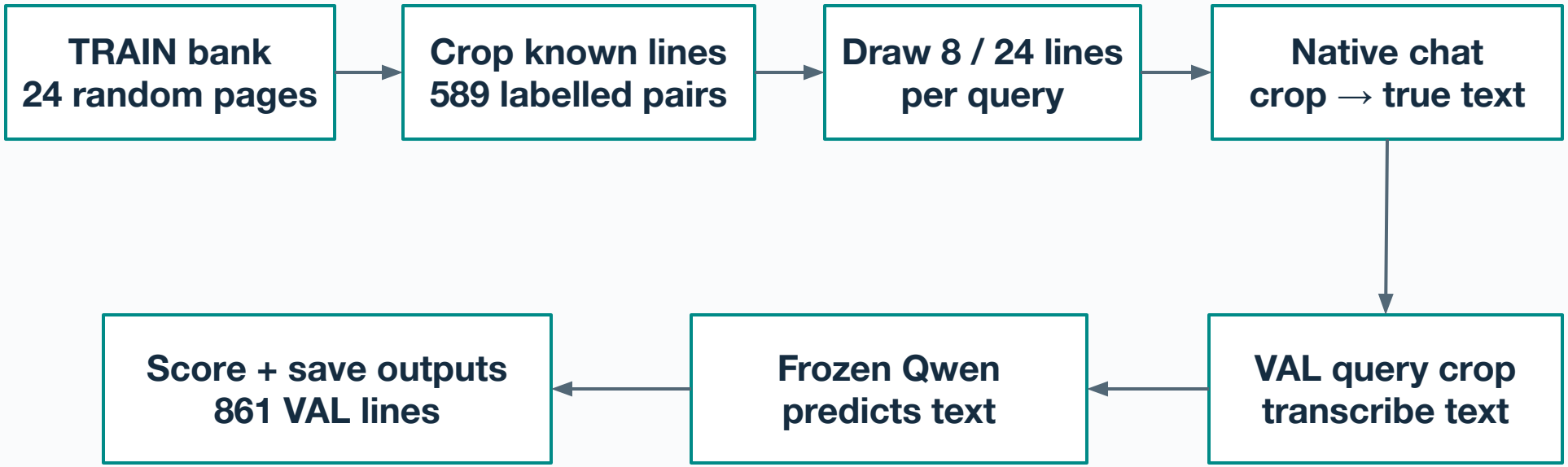
At 24 pages, training exposure differs: random 589 lines; k-center 501.

Small gaps also mix data selection with unseeded adapter initialization; repeatability is not established.

EXP-013 · CER/WER are error ratios, lower is better · *0 pages reused from EXP-012

8 diagnostic lines
checked at 0/8/24 shots

ICL supplies examples before one query crop



Fresh chat for every query.
The query crop is always present; validation labels never enter the prompt.

24-shot test: 25 images; 3,999-token allowance fits · support draws are reproducible

One prompt = demonstrations + one query

Actual chat input · k = 0 / 8 / 24

USER	[TRAIN crop 1] + instruction P
ASSISTANT	true TRAIN transcript 1
	⋮ repeat for k crop–transcript pairs
USER	[VAL query crop] + instruction P
ASSISTANT	**model generates this transcript**

Demo answers are supplied; only the final answer is generated.

Context = text + image tokens

Shots	Images	Max input tokens
8	9	1,826
24	25	4,358

0-shot: one query image, no examples.

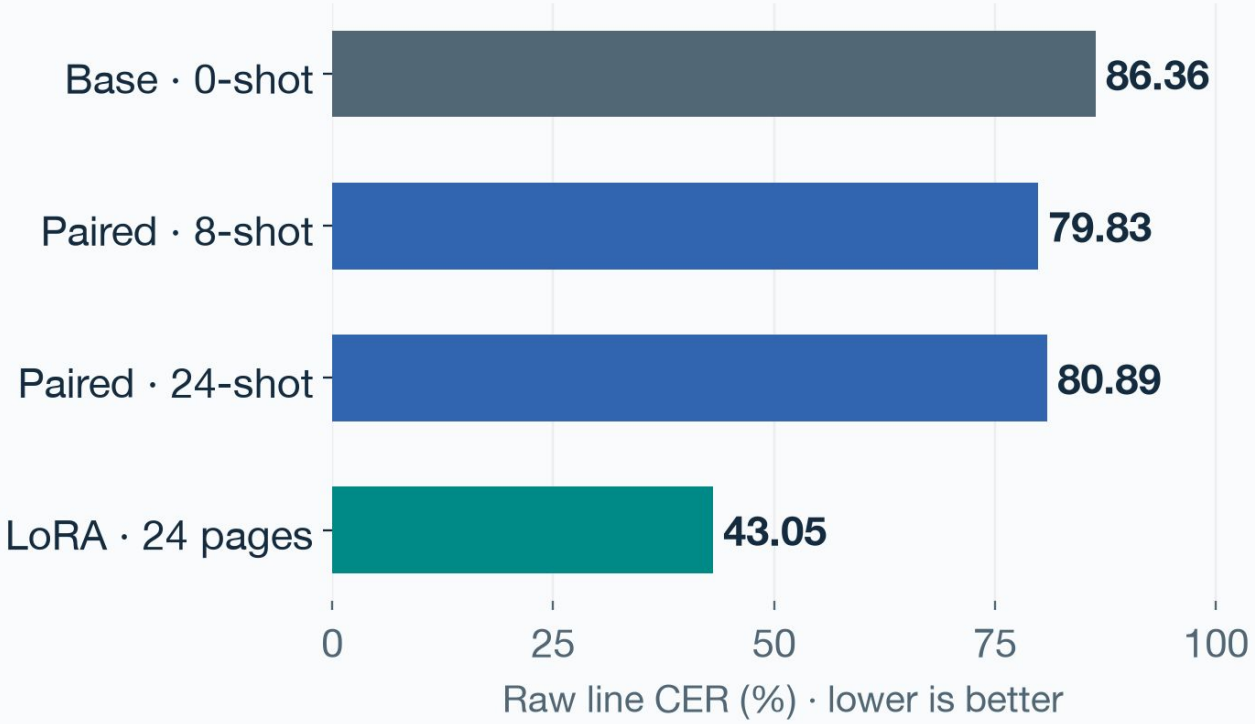
Qwen configured limit: 262,144
Largest input + 64 output: 4,422

Image tokens depend on crop resolution.
GPU fit must also be checked.

P: “Transcribe only the text in this cropped line. Return only the transcription.”

No system turn · actual crop pixels · VAL truth is used afterwards for CER / WER

ICL improves Qwen, but remains far behind LoRA



Mode	Shots	CER %	WER %
Base	0	86.36	117.46
Paired	8	79.83	112.11
Paired	24	80.89	112.22
LoRA	0*	43.05	83.48

More examples do not consistently improve OCR.

*LoRA uses all 589 bank lines from 24 TRAIN pages; ICL exposes 8/24 per query.

Same bank, unequal information exposure: this is not a compute-matched learning comparison.

EXP-015/016 raw line metrics · zero-shot: reused EXP-012 · LoRA: reused EXP-013

Do support images add useful information?

Hold the query fixed

Same support IDs and transcripts.
Same ordering and decoding.
Change only the support imagery.

Three matched conditions

Paired: crop with its transcript.
Text-only: remove support crops.
Mismatch: rotate support crops.

PAIRED	TEXT-ONLY	MISMATCHED
image + its transcript	same transcripts	image + wrong transcript

Every condition still sees the actual VAL query crop.
No training, adapters or validation transcripts enter inference.

Interpretation target: information used in context, not proof of a specific attention mechanism.

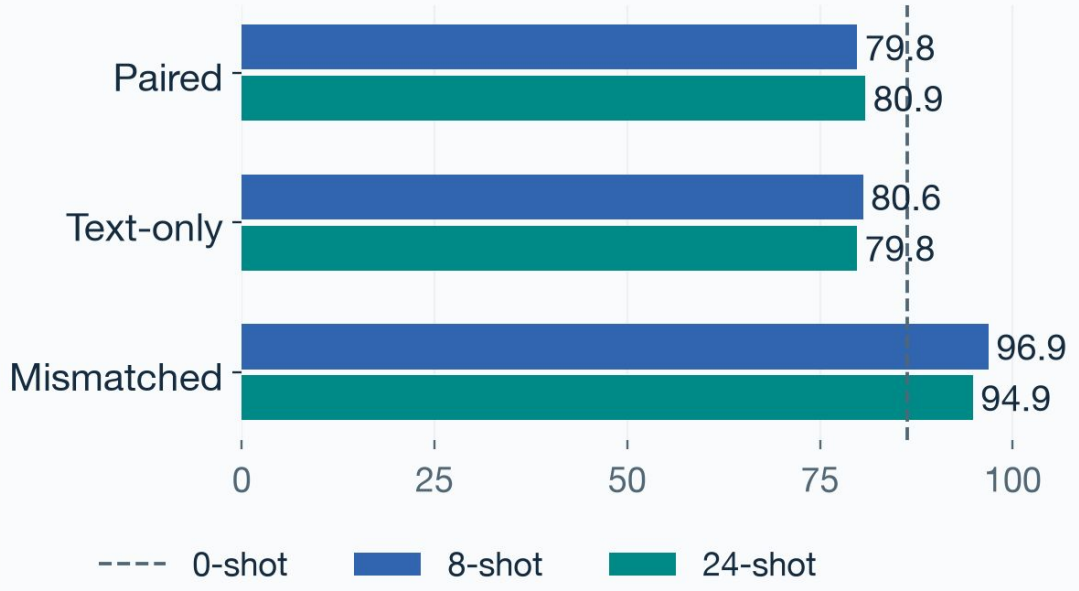
Controls motivated by Chen et al. (WACV 2025) and Bertini Baldassini et al. (CVPRW 2024)

Text-only ICL is close; mismatched pairs hurt

Matched controls · error %

Mode	Shots	CER %	WER %	Caps
Paired	8	79.83	112.11	2
Paired	24	80.89	112.22	3
Text-only	8	80.60	113.01	1
Text-only	24	79.81	109.87	0
Mismatch	8	96.89	129.31	2
Mismatch	24	94.87	126.36	5

Raw CER % ↓



Paired 8 vs text-only 24 differs by only four character edits.

Wrong script nearly disappears even for harmful mismatched context.

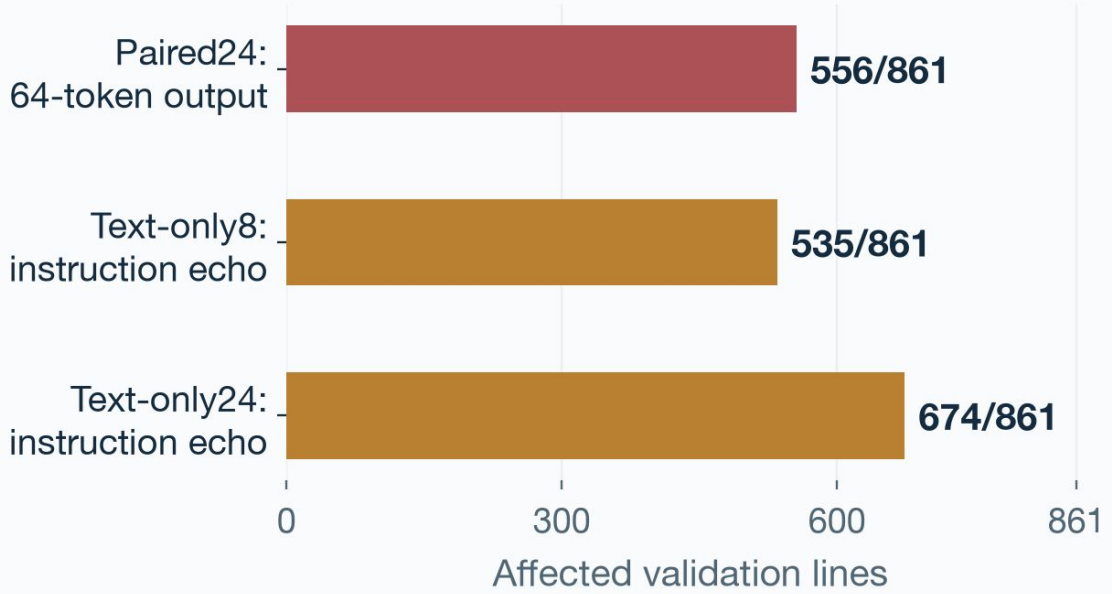
Closeness in one pilot is not statistical equivalence; script correction is not accurate transcription.

The Qwen chat protocol does not transfer cleanly

Full cross-model results · %

Mode	Q CER	Q WER	L CER	L WER
Zero	86.36	117.46	152.35	200.60
Paired 8	79.83	112.11	332.37	376.11
Paired 24	80.89	112.22	359.74	380.90
Text 8	80.60	113.01	635.00	693.72
Text 24	79.81	109.87	422.42	470.84
Mismatch 8	96.89	129.31	296.44	341.42
Mismatch 24	94.87	126.36	347.28	368.19

LightOn output behavior



Observed: instruction echoes, 64-token outputs and 145 empty zero-shot lines.

A prompt-free OCR base model is not automatically a multi-turn ICL model.

This tests protocol compatibility, not a calibrated native handwriting benchmark or a universal model ranking.

Retaining failed outputs changes the interpretation

Input isolation and triggering verified; capped predictions must remain visible.

Why keep repetition penalty 1.1?

VAL8: penalty 1.1 → 1.0
Mean raw CER: 83.50% → 98.52%.
Caps: 2 → 25 across 0/8/24 shots.
The bounded test rejected the change.

Why replace the headline metric?

Old scoring blanked capped strings.
LightOn zero → paired24 looked better:
old CER 89.91% → 83.82%;
raw CER 152.35% → 359.74%.

Current standard: corpus CER/WER on raw per-line predictions; NFC only.

CER = character edits / reference characters; WER uses word edits / reference words.

No cap blanking or silent reruns. EXP-013 curves keep their labelled historical metric.

Audit: EXP-015–017 receipts, prompt reconstruction, saved outputs and independent metric checks

The READ result is difficult in the cited ICL study too

READ / ICFHR16 · CER %

Study	0-shot	8-shot	16-shot	24-shot
Paper mean	80.95	81.96	78.78	—
Our raw	86.36	79.83	—	80.89

Paper: Qwen8B; test; 30 draws/query.
Ours: Qwen4B; VAL42; one draw/query.
Exact paper prompt / token limits unreported.

Why our controls matter

Chen: demonstration text can dominate; the query image still matters.

Bertini Baldassini: pairing and order effects depend on task and model.

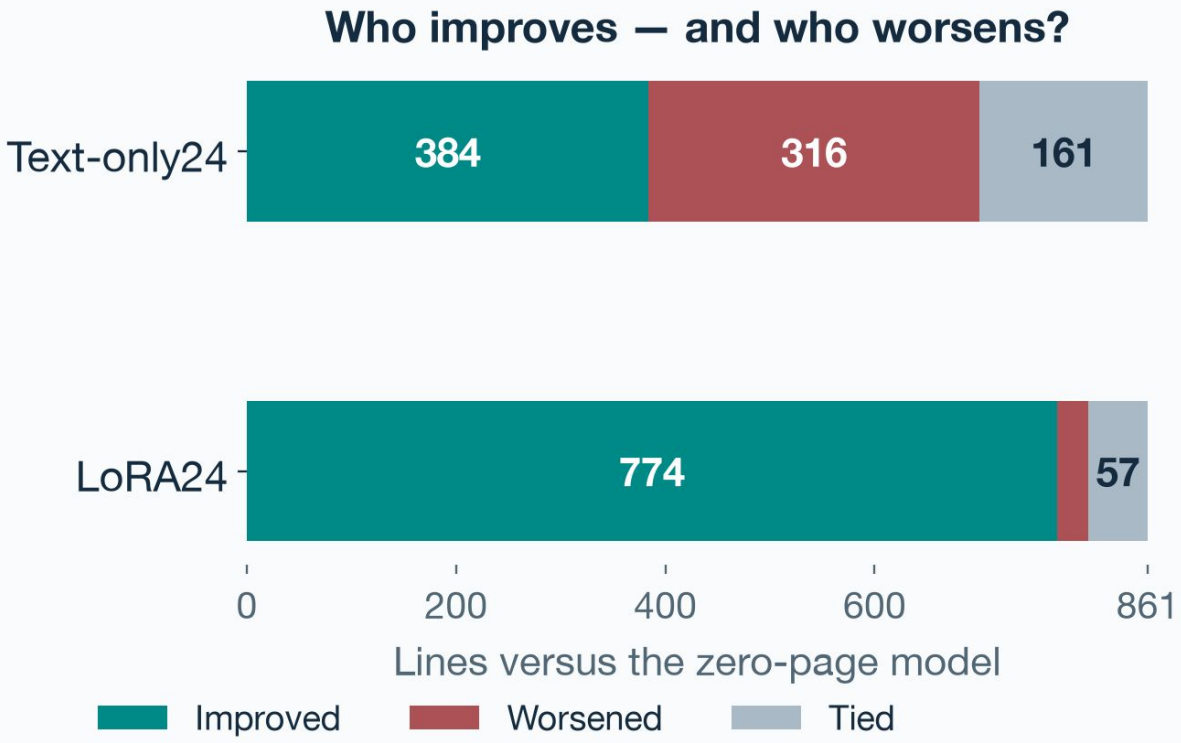
The paper shows limited average READ gains; ICL is not uniformly strong.

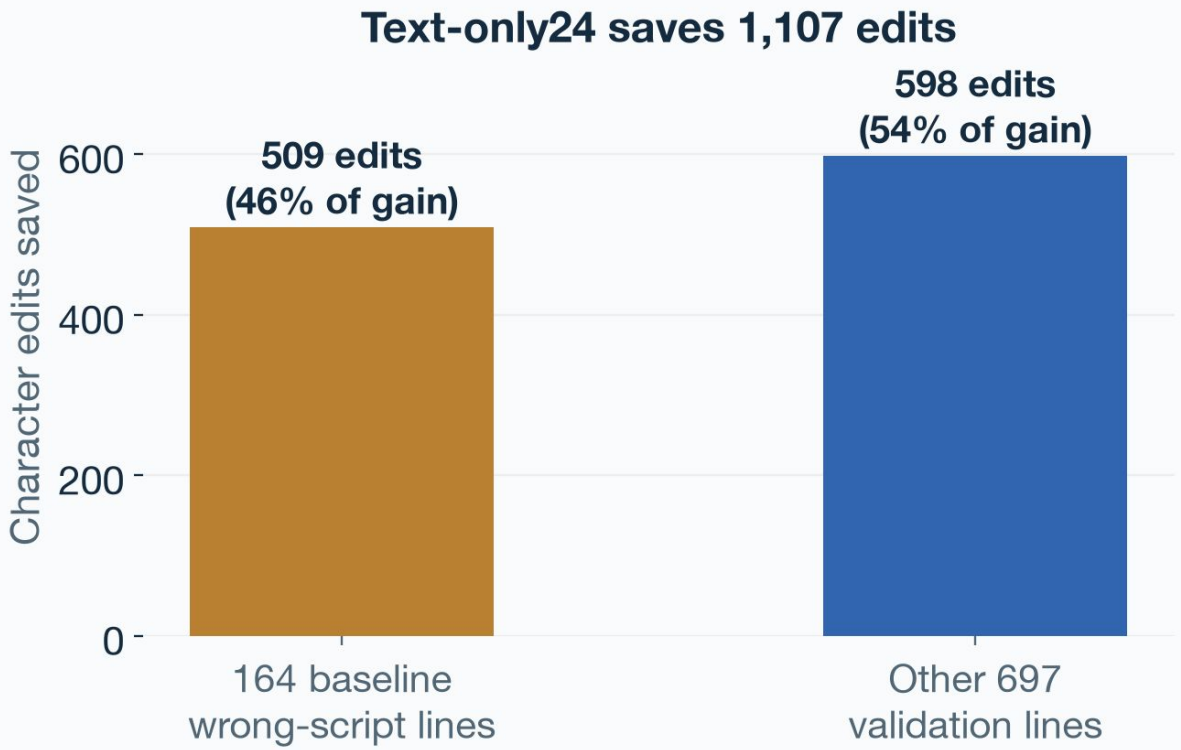
Its best-context 16-shot CER is 55.61% — chosen with the true answer.

Oracle sampling is not deployable. These studies motivate controls, not a causal explanation of Qwen.

Ayllon et al. 2026 preprint · Chen et al. WACV 2025 · Bertini Baldassini et al. CVPRW 2024

Text context helps unevenly across lines





Text-only24 improves 384 lines but worsens 316; LoRA improves 774, worsens 30.

Wrong script nearly vanishes, yet CER is still 86.4% in that baseline subgroup.

Language priors help, but can also rewrite the source

Local spelling priors, ordered text and domain adaptation are different mechanisms.

Target-domain text can help

Meyer et al.: character 5-gram injection.
IAM domain-shift CER 18.0% → 10.1%.
A specially trained recognizer consumes the text-derived prior at inference.

Fluency can alter historical spelling

Kanerva et al.: normalized and literal evaluation can give opposite conclusions.
Vesalainen et al.: Qwen sometimes changes “Antient” into “Ancient”.

Our unresolved question: language identification, vocabulary or ordered text?

Measure literal OCR errors; better-looking language is not sufficient evidence.

Different corpora and architectures: these findings guide controls, not a matched benchmark.

Meyer et al. 2026 preprint · Kanerva et al. RESOURCEFUL 2025 · Vesalainen et al. 2026 preprint

Next: isolate the useful linguistic information

First diagnostic: four arms

A: current zero-shot prompt.
B: add a verified collection-language cue.
C: B + 8 intact TRAIN transcripts.
D: B + the same word-shuffled text.

What each comparison tests

B–A: explicit language priming.
C–B: transcript information.
C–D: benefit of ordered text.
Keep the real query crop in every arm.



Fix bank/draws, decoding and raw metrics; TRAIN-only context.
For adaptation: match label exposure, modules and initialization.

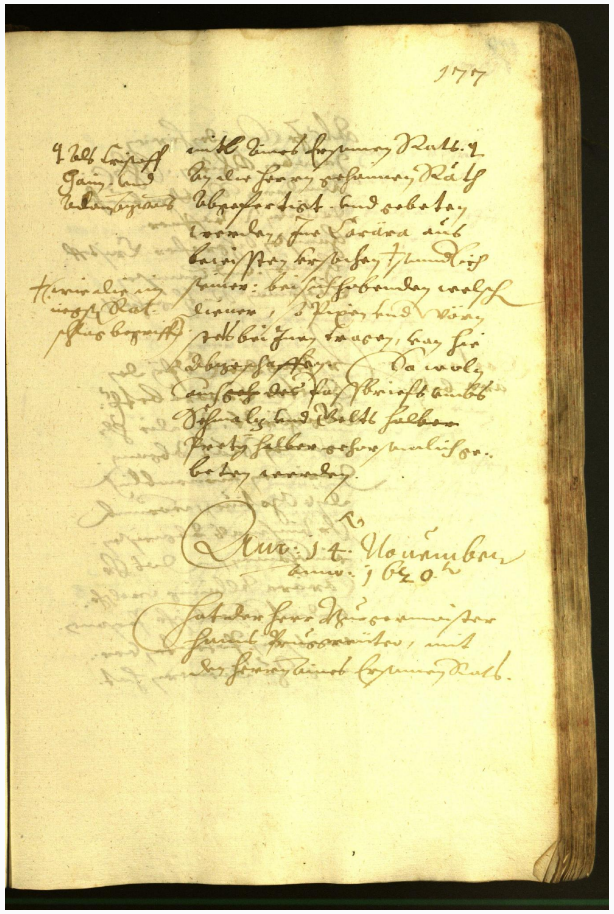
Contribution sought: when linguistic information helps literal historical OCR, and when it hurts.

QIAO Bowen - FYP RESEARCH PROGRESS · 27 SEPTEMBER 2026

Active learning for historical OCR

Can better page selection reduce the labelled data needed to adapt small vision-language models?

Dataset and the unit of annotation



Historical German handwriting

Official split	Pages	Text lines
TRAIN pool	350	8,367
VAL	50	1,043

Select one TRAIN page → reveal every line transcript

Existing true labels simulate annotation.
Model scoring sees images; VAL truth stays with the evaluator.

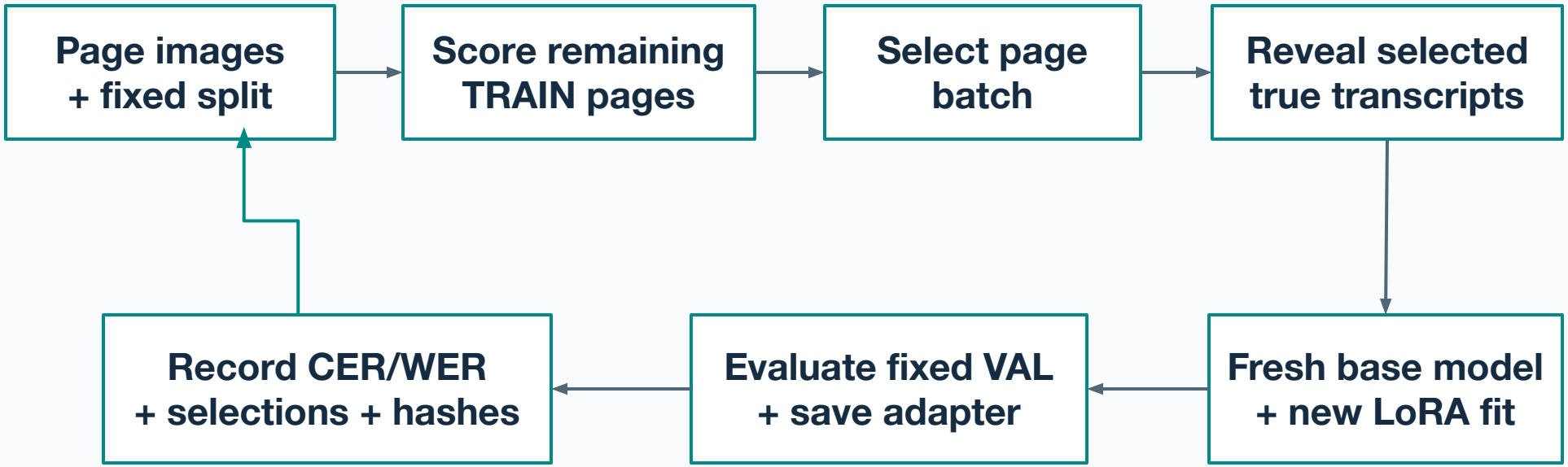
8 VAL pages for recipe checks · 42 fixed VAL pages for reported curves

Source: READ2016 / Bozen v1.2.0 · Toselli et al. · CC BY 4.0

ENGINEERING STRUCTURE

The active-learning pipeline

Original model evaluated at 0 labelled pages



Each round fits all acquired labels;

New LoRA adapter from the same pretrained model and train it on *all pages selected so far*. Ash & Adams, 2020; Das et al., 2023.

TRAIN oracle ≠ VAL evaluator · Random skips scoring · Stop after 3 fixed rounds

Why the first joint OCR recipe was paused



What we established

Training, checkpoint reload and evaluation eventually completed. We also tried stronger fitting and simpler box/text output.

What prevented a useful comparison

Outputs stayed invalid, repetitive or capped. CER stayed near 1; line-box F1 stayed at 0. Simplifying JSON alone did not fix recognition.

Research decision: isolate OCR transcription, NO text detection

Poor usable OCR masks acquisition effects. This rejects the tested recipe, not all joint OCR.

Evidence: grouped reader reports in results/EXP-001-002.md, EXP-003-005.md, EXP-006-008.md

Can historical page transcription learn?

Question: does plain page text learn before we compare selectors?

EXP-009 · LightOnOCR-2-1B-base

OCR-finetuned VLM

Built for page-image transcription; the author recommends this base variant for domain fine-tuning.

EXP-010 · Qwen3-VL-4B-Instruct

General-purpose VLM

Advertises document OCR capability. Keeping Qwen tests whether simplifying the output task can unlock learning.

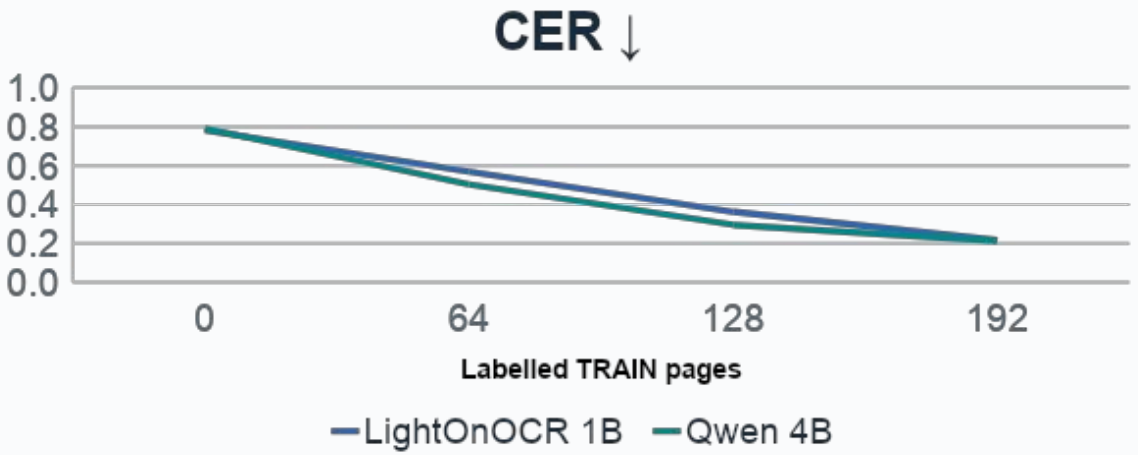
Shared numeric recipe: random 0 → 64 → 128 → 192 pages; seed 824

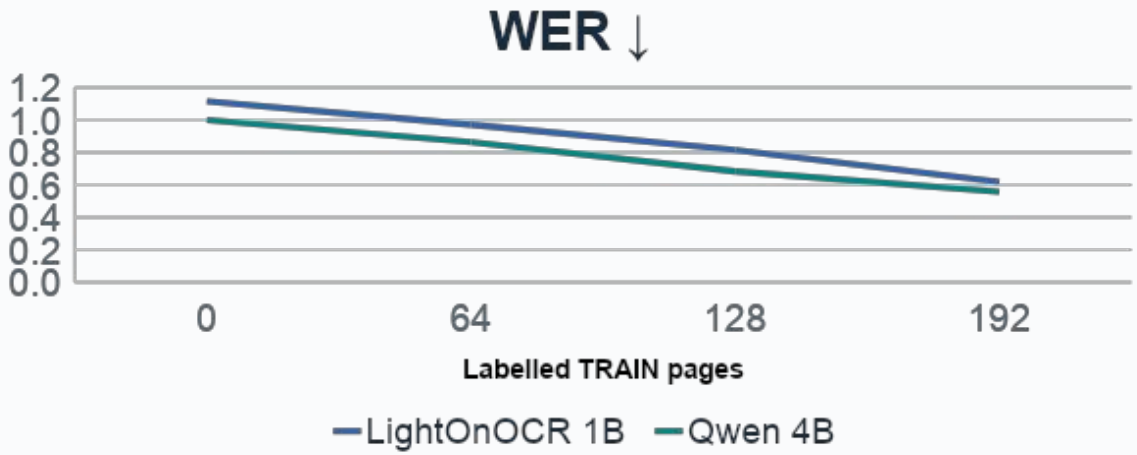
3 epochs per fresh cumulative fit · LoRA r8 / α16 · LR 10^{-4} · HF Transformers + PEFT

Specialist versus general VLM is an engineering comparison; processors and prompts differ.

Model sources: LightOnOCR base card / report · Qwen3-VL-4B-Instruct card / report

Both models learn under random selection





Pages	LightOn CER	LightOn WER	Qwen CER	Qwen WER
0	0.7812	1.1155	0.7902	1.0019
64	0.5683	0.9720	0.5025	0.8643
128	0.3625	0.8164	0.2946	0.6841
192	0.2161	0.6179	0.2112	0.5589

~72–73% relative CER reduction at 192 pages; substantial word errors remain.

Same fixed VAL42 · CER/WER are edit counts divided by reference length; lower is better

Entropy and confidence estimate OCR uncertainty

p□(v): raw next-token probability · y□: generated token · T: tokens · V: vocabulary size

Entropy

ENTROPY

$$H(\text{line}) = -\frac{\sum_{t=1}^{T}\sum_{v=1}^{V} p_t(v)\log p_t(v)}{T\log V}$$

Select the pages with the highest uncertainty.

Least confidence

LEAST CONFIDENCE

$$C(\text{line}) = \exp\left[\frac{1}{T}\sum_{t=1}^{T}\log p_t(y_t)\right]$$

Select the pages with the lowest token confidence.

Random control

Seeded uniform page selection without replacement; no model score.

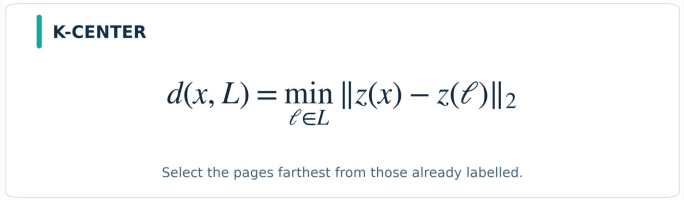
Random sampling can sometimes outperform entropy and least-confidence selection when uncertainty highlights unhelpful examples (Jedoui et al., 2019); broader benchmarks also find inconsistent gains over random (Munjal et al., 2022).

EXP-011: one page sequence. EXP-013: average 3 fixed line scores into each page score.

Uncertainty motivation: Settles (2009), §3.1 · formulas below are our generated-token adaptation

K-center expands frozen visual coverage

z: frozen normalized page feature · L: acquired pages · U: remaining pages



Choose $x^* = \text{argmax}_{x \in U} d(x, L)$; add x^* to L; update distances; repeat.



EXP-011: begin from the shared 64 selected pages.

EXP-013: with L empty, first choose the page nearest the pool's mean feature.

Why: cover different inputs without relying on text confidence; visual distance may miss OCR difficulty.

Coverage motivation: Sener & Savarese, ICLR 2018 · greedy Euclidean coverage in frozen features

Whole-page selection after a shared warm start



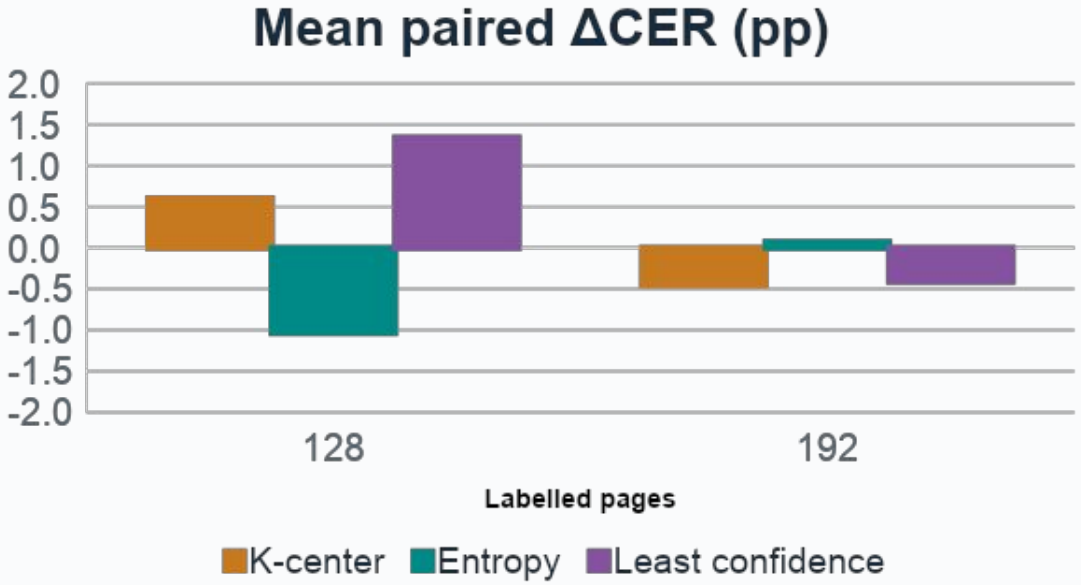
Purpose: test which remaining pages help, after the same initial supervision.

Strategy	128 CER	128 WER	192 CER	192 WER
Random	0.2856	0.6720	0.2096	0.5550
K-center	0.2917	0.6865	0.2051	0.5554
Entropy	0.2753	0.6709	0.2104	0.5533
Least conf.	0.2991	0.6834	0.2055	0.5543

All arms: 0-page CER/WER 0.7902 / 1.0019; mean at 64: 0.4996 / 0.8637.

Entropy leads mean CER at 128, but the later ranking changes; paired gaps tell the fuller story.

Whole-page selection gains vary across seeds



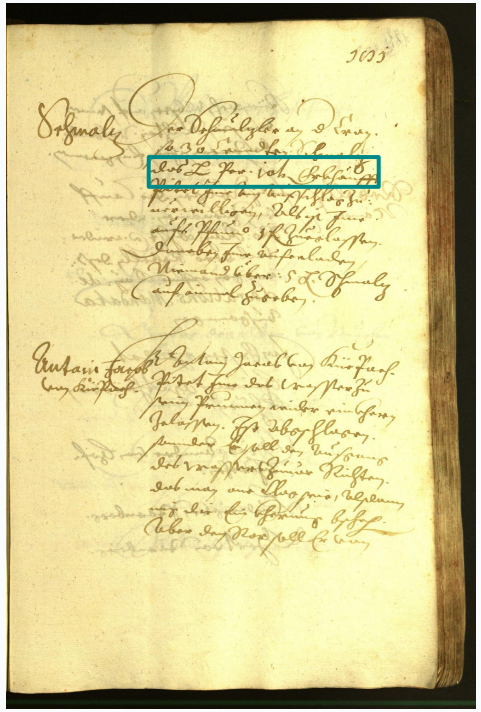
Method / pages	ΔCER ± SD	ΔWER ± SD
K-center 128	+0.60 ± 1.07	+1.45 ± 0.21
K-center 192	–0.45 ± 0.85	+0.04 ± 1.00
Entropy 128	–1.03 ± 0.69	–0.11 ± 0.82
Entropy 192	+0.08 ± 2.18	–0.17 ± 1.44
Least conf. 128	+1.34 ± 3.85	+1.13 ± 1.58
Least conf. 192	–0.41 ± 2.48	–0.07 ± 1.31

No selector consistently wins both budgets across CER and WER.

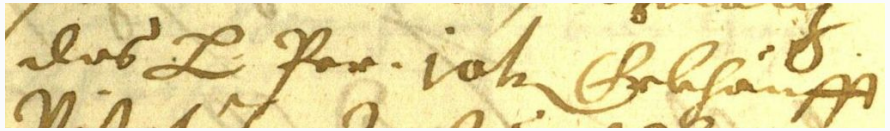
Seed 824: 26 pool outputs hit the cap yet had low entropy / high confidence.
Uncertainty can miss repetitive failures; uncapped-page checks changed some rankings.

Δ = selector – same-seed random · percentage points · negative values favour the selector

Known boxes isolate text recognition



Whole page



Supplied line box → Qwen transcribes only this crop

Short text outputs in source reading order

Why this helps

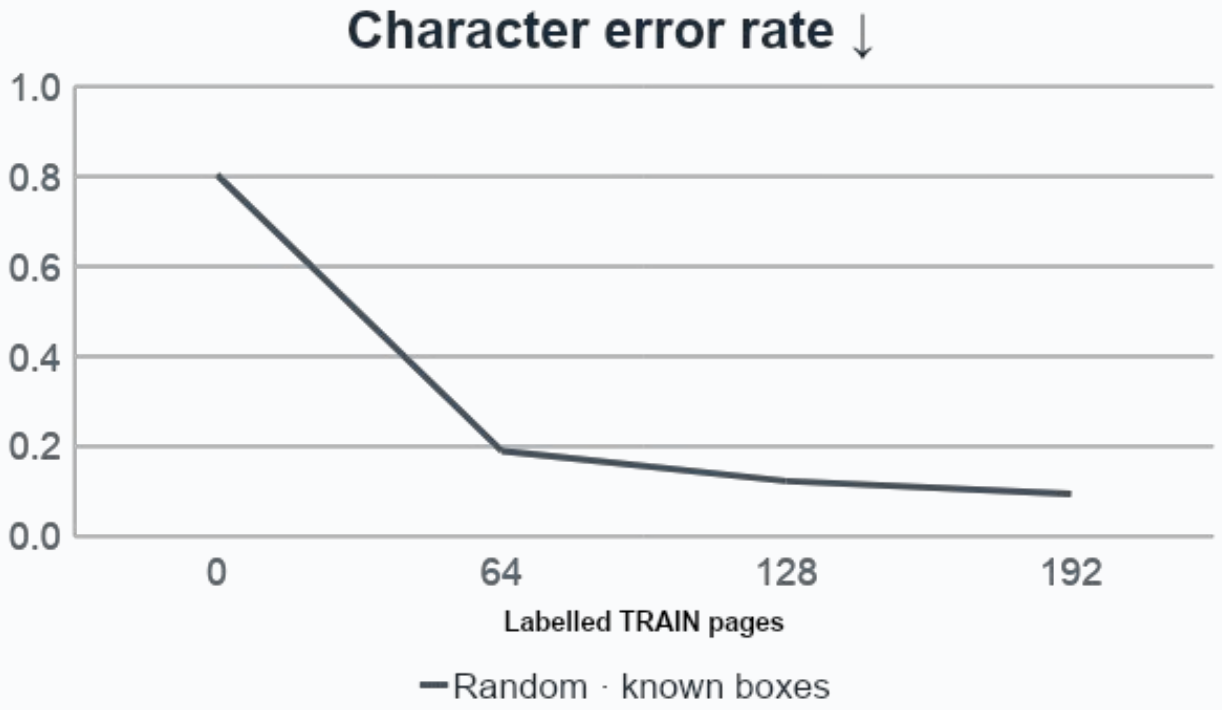
Separate recognition from localization.

Reduce long-output failures.

Study selection with usable OCR.

This is a different task and recipe; its CER is not a fair head-to-head gain over whole-page OCR.

Random selection establishes a known-box baseline



Pages	Train lines	CER	WER
0	0	0.804	1.087
64	1,537	0.190	0.581
128	3,037	0.123	0.452
192	4,567	0.094	0.370

The largest drop comes with the first 64 pages

1 epoch per fresh fit · 3 fixed rounds · capped VAL regions: 2 → 0 → 0 → 0

Establishes adaptation and a control curve; no model-informed selection comparison yet.

Source: results/EXP-012.md · same 42 VAL pages / 861 lines at every checkpoint

Strategy-first selection at smaller page budgets

Why smaller budgets?

EXP-012's largest gain came early.
24-page batches leave more headroom
to test the earliest selection effects.

Why remove the warm start?

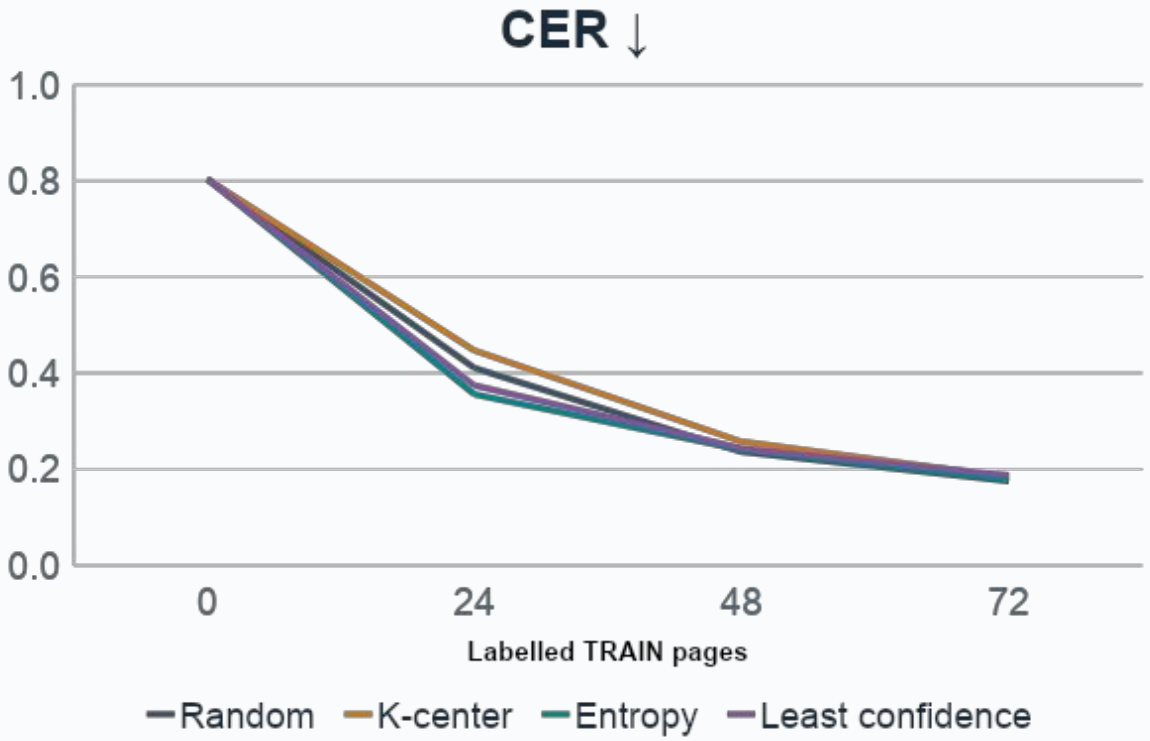
EXP-011 tested additions after 64 labels.
Now each strategy chooses its own
first 24 from the original model.

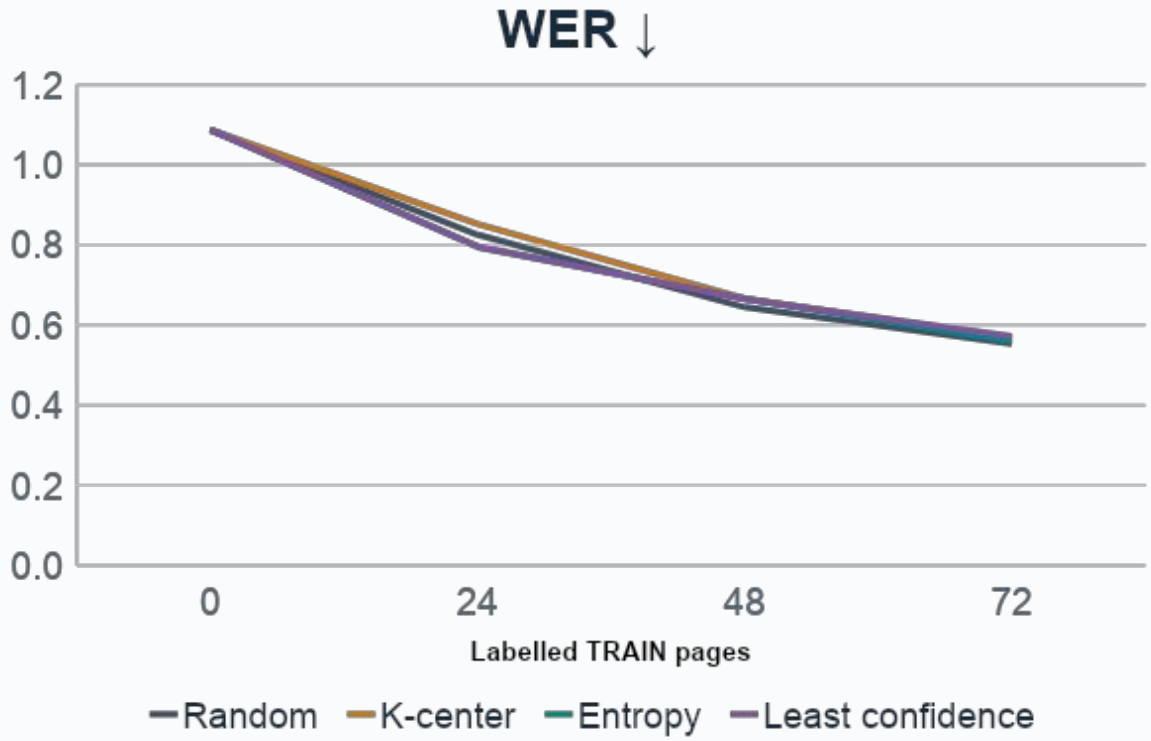


Score lines at ¼, ½, ¾ of reading order; selection reveals all page labels.
Later uncertainty uses its own adapter; k-center features stay frozen.

0-page VAL42 baseline reused from EXP-012 after identity checks and metric recomputation.

Uncertainty leads early; random leads later





Early lead: entropy reduces CER versus random by 5.45 percentage points at 24.

The lead is not sustained; this is one pilot seed, not proven annotation savings.

Same VAL42 / 861 supplied lines · original-model zero baseline reused from EXP-012

RESULTS TABLE · RATES, NOT PERCENTAGES · LOWER IS BETTER

EXP-013: CER and WER at every page budget

Character error rate (CER)

Strategy	0*	24	48	72
Random	0.8036	0.4104	0.2357	0.1750
K-center	0.8036	0.4469	0.2574	0.1838
Entropy	0.8036	0.3559	0.2400	0.1781
Least conf.	0.8036	0.3740	0.2439	0.1881

Word error rate (WER)

Strategy	0*	24	48	72
Random	1.0871	0.8254	0.6445	0.5529
K-center	1.0871	0.8523	0.6662	0.5664
Entropy	1.0871	0.7951	0.6665	0.5593
Least conf.	1.0871	0.7963	0.6650	0.5716

24 pages: entropy leads both metrics; confidence also improves over random.

72 pages: entropy is close, but random has the lowest CER and WER.

*0 is reused, not a new run. Capped VAL lines: 2/861 at 0; 0/861 at every trained checkpoint.

Values rounded to 4 decimals; exact metrics and receipts are preserved in results/EXP-013.md

Uncertainty strategies select overlapping pages

Cumulative page intersection

Strategy pair	24	48	72
Random / K-center	2	6	13
Random / Entropy	2	6	14
Random / Confidence	2	5	17
K-center / Entropy	1	5	16
K-center / Confidence	1	9	20
Entropy / Confidence	16	24	37

Cumulative training lines

Strategy	24	48	72
Random	589	1,155	1,725
K-center	501	1,071	1,658
Entropy	580	1,160	1,713
Least conf.	584	1,116	1,659

Entropy/confidence share 37 of 72 pages. Different scores can yield similar batches.

Equal page budgets do not mean equal transcript volume or equal human annotation effort.

Pair overlap = shared acquired TRAIN pages · line counts = all labels revealed by those pages

Current findings and the next research test

Established

Domain fine-tuning improves recognition.

Both page and crop pipelines run.

Random is a necessary strong control.

Still unproven

Repeatable annotation savings (beat random selection).

Independent-document/dataset generalization.

Strategies for Active-Learning.

EXP-013: one seed; LoRA initialization was not explicitly seeded. Official validation only.

Next: Consider combination of strategies for ActiveLearning, Faithful Linguistic Prior

Hybrid selection: Combine uncertainty and visual diversity; test whether it beats random. (Ash et al., BADGE, 2020)
Faithfulness direction: Use linguistic context, but check that fluent text remains true to the image. (Li et al., TrOCR, 2023; Lee et al., 2026)

Sources that motivate the research design

Settles · Active Learning Literature Survey (2009)

Classical uncertainty sampling; our token/page scoring is an adaptation.

Sener & Savarese · ICLR2018

Core-set coverage motivates the greedy k-center baseline.

Munjal et al. · CVPR2022

Strong random controls and stochasticity-sensitive evaluation.

Aradillas, Murillo-Fuentes & Olmos · 2020 preprint

Historical handwriting recognition with few labelled lines.

Bai et al. · Qwen3-VL (2025) / **Taghadouini et al. · LightOnOCR (2026)**

General VLM and compact OCR-specialist model context.

Toselli et al. · READ2016 / Bozen v1.2.0

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Experiment results index: grouped EXP-001–008 and individual EXP-009–013 reports